

Saber Ayatollahi, CFA, MSc

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SUMMARY

Senior Analyst (Senior Associate), Portfolio Analytics candidate with ~7 years of multi-asset investment and risk analytics for pension and institutional portfolios. Ran VaR/CVaR portfolio optimization, risk decomposition, and currency, inflation, and duration/leverage overlay analysis at Ortec Finance; now holds delegated sign-off on valuation and sensitivity outputs for \$5-25bn multi-asset portfolios at Moody's Analytics. Re-engineers manual workflows into production Python (pandas, NumPy, SciPy) and PostgreSQL analytics pipelines. CFA charter, dual MSc.

CORE SKILLS

Multi-Asset Portfolio Analytics • Portfolio Optimization (VaR / CVaR) • Risk Decomposition & Attribution • Derivatives Valuation (Rates, FX, Inflation) • Quantitative Modelling & Monte Carlo Simulation • Stochastic Scenario Generation & Stress Testing • Python (pandas, NumPy, SciPy) • SQL / PostgreSQL Analytics Pipelines • Git & CI/CD Workflows (professional-context) • Currency, Inflation & Duration Hedging Analysis • Strategic & Tactical Asset Allocation (SAA / TAA) • Requirements Translation for Investment Teams

EXPERIENCE

Moody's Analytics

Asst. Director, Modelling Services

Toronto, May 2023 - Present

Multi-Asset Portfolio Analytics & Cross-Asset Valuation

- Hold delegated sign-off authority on valuation, sensitivity, and portfolio risk outputs delivered to multi-asset institutional portfolios of \$5-25bn per engagement (~\$50bn+ cumulative), independently reviewing curve construction, spread calibration, and cross-asset interactions prior to production release.
- Validate derivatives pricing outputs across rates, FX, and inflation, confirming consistency of sensitivities and scenario impacts at portfolio-level aggregates.
- Review aggregation logic converting security-level exposures into portfolio-level risk metrics feeding downstream analytics and reporting.
- Serve on the model governance committee, contributing to methodology review, documentation and benchmarking standards, and model-performance assessment for client-delivered analytics.

Quantitative Modelling & Scenario Analytics

- Led design and implementation of an enterprise multi-asset cash flow projection engine supporting base, stress, and reverse-stress scenarios across investment horizons.
- Architected configurable time-bucketed projection analytics (T+1 through multi-year) used for funding, liquidity, and asset-allocation decisions.
- Oversee duration and interest-rate exposure analysis under parallel and non-parallel curve shocks, incorporating spread calibration across fixed income holdings feeding portfolio-level risk aggregates.

Data Pipelines & AI-Assisted Engineering

- Re-engineered manual spreadsheet workflows into scalable Python (pandas, NumPy, SciPy) and PostgreSQL analytics pipelines with embedded logging, validation, and auditability controls, running shadow cycles before cutover.
- Built agentic AI development workflows (Claude Code, Cursor IDE) automating code generation, validation scaffolding, anomaly detection, and documentation, cutting development cycle time an estimated 30-40% on comparable modules.

Partnering with Investment Teams

- Act as escalation point for client-critical analytical issues, holding releases where outputs lack mathematical or economic defensibility and walking Heads of Risk and investment stakeholders through remediation.
- Translate quantitative output into decision-ready narratives on exposure, scenario impacts, and portfolio sensitivities for senior investment stakeholders.

Client Service Specialist

Toronto, May 2022 - May 2023

- Onboarded U.S. and Canadian pension funds, asset managers, and consultants onto PFaroe DB (risk analytics) and PFaroe PM (portfolio management), scoping investment-team requirements into structured Product Owner requests for development.
- Migrated all assigned client accounts from the legacy Calypso platform to PFaroe PM, validating model outputs post-deployment.

Ernst & Young (EY)

Senior Consultant, IFRS & Risk Transformation

Toronto, Sep 2021 - Apr 2022

- Delivered IFRS 17 and IFRS 9 technology-transformation projects for Canadian and international insurers, coordinating actuarial, finance, IT, and risk teams on data sourcing, milestone delivery, and governance documentation.

Ortec Finance Canada

Consultant, ALM & Risk

Toronto, Feb 2019 - Sep 2021

- Performed asset-only and asset-liability (surplus) portfolio optimization on VaR and CVaR using Ortec's GLASS platform, with risk decomposition, contribution-to-risk budgeting, and near-optimal frontier analysis to test the robustness of allocation recommendations.
- Built and interpreted stochastic economic scenario generators (Monte Carlo) assessing interest-rate risk and funding volatility under base-case and stressed regimes.
- Advised institutional clients on Strategic Asset Allocation and Tactical Asset Allocation trade-offs through stress testing and sensitivity analysis.
- Conducted cash flow, liquidity, and hedging analysis across interest-rate, inflation, and currency risk for international institutional clients.
- Led the ALM study for a three-plan university pension merger on a team of three, developing and validating the model and evaluating duration, currency, inflation, and leverage overlay strategies.
- Presented study findings and allocation recommendations to investment committees at Canadian pension-fund clients on-site.

EDUCATION & CREDENTIALS

- Chartered Financial Analyst (CFA) - CFA Institute, 2024
- MSc, Financial Modelling - Western University, 2018
- MSc, Chemical Engineering (Full Scholarship) - Western University, 2016